EEEN30160 – Biomedical Signal Processing

Assignment 4

Oisín Hogan

19355123

Contents

Problem 1: 3
Code: 3
Output ..... **Error! Bookmark not defined.**
Discussion 3
a) ..... **Error! Bookmark not defined.**
b) ..... **Error! Bookmark not defined.**
c) ..... **Error! Bookmark not defined.**
Problem 2: 4
Code: 4
Outputs: 5
Discussion: 6
a) 6
b) ..... **Error! Bookmark not defined.**
c) ..... **Error! Bookmark not defined.**
d) ..... **Error! Bookmark not defined.**
Problem 3: 6
Code: 7
Output: 9
Discussion 9
Problem 4: ..... **Error! Bookmark not defined.**
Code: ..... **Error! Bookmark not defined.**
Output: ..... **Error! Bookmark not defined.**
Discussion ..... **Error! Bookmark not defined.**

Problem 1:

Code:

```
clc
clear
close all% clear the command windows, workspace and figures

Fs = 2000;% set the sampling frequency
Fn = Fs/2;% set the nyquist
Ws = 100/Fn;Wp = 350/Fn;% calculate the Stop and Pass frequencies
Rp = 1; Rs = 40;% calculate the stop and pass ripples
%Calculate the orders and frequencies of the different filters
[Nb,Wcb]=buttord(Wp,Ws, Rp,Rs) %Butterworth filter
[Nc1,Wcc1]=cheb1ord(Wp,Ws,Rp,Rs) %Chebyshev Type-I
[Nc2,Wcc2]=cheb2ord(Wp,Ws,Rp,Rs) %Chebyshev Type-II
[Ne,Wce]=ellipord(Wp,Ws,Rp,Rs) %Elliptic
```

Outputs

```
Nb =      4

Wcb =     0.2956

Nc1 =      3

Wcc1 =     0.3500

Nc2 =      3

Wcc2 =     0.1000

Ne = 3

Wce =     0.3500
```

Discussion

1. which filter would be better to use if minimizing the number of operations is one of the aims of your analysis?

Either the Elliptic, Chebyshev1 or Chebyshev2 filters would be the valid to minimise the number of operations as they had a lower exponent, 3, than the Butterworth filter, which was 4.

2. which filter would be better to use if a steep roll-off is a key requirement?

The elliptical filter would provide the steepest roll off

Problem 2:

Code:

```
clear
clc
close all% clear the command windows, workspace and figures
load("G:\My Drive\engineering_work\Biosigs\Lab 4\NoisyECG.mat")%Load the ECG file

Fs = 500;% sampling frequency
Fn = Fs/2;%Nyquist Frequency
Ampf = 500; % the amplification factor
Ecg = (ecg/500)*1000;% Find the original ecg signal in mV
len = length(ecg); % number of samples
%a)
tiledlayout(3,2)% Tiled Layout
nexttile
plot(Ecg) %plot original signal
xticks([]) % remove 'x' numbers, as they will be just on the bottom graphs
ylabel('Amplitude (mV)')% y labels
title('ECG Signal') % title

ecgf = abs(fft(Ecg));
ecgf = fftshift(ecgf);% fourrier transform of the signal
f = (-len/2:(len-1)/2)*Fs/len; %frequency range of the signal

nexttile
plot(f, ecgf)%plot the frequency domain of the signal
xticks([])
ylabel('Magnitude (dB)')
title('ECG Signal')
xlim([0, Fs/2])
%b)
Fc = 50;% choose frequency of notch
wo = Fc/Fn; % calcualte the adjusted frequency
bw = wo/30; % calculate bw
[b,a] = iirnotch(wo,bw); % get constants, 'a' and 'b'
ecg2 = filtfilt(b, a, Ecg); % Filter ecg
ecg2f = abs(fft(ecg2)); %get the frequencies of filtered ecg
ecg2f = fftshift(ecg2f);
nexttile
plot(ecg2)% plot filtered ecg in time and frequency domain
xticks([])
ylabel('Amplitude(mV)')
title('Filtered ECG Signal')
nexttile
plot(f, ecg2f)
xticks([])
ylabel('Magnitude (dB)')
xlim([0, Fs/2])
title('Filtered ECG Signal')
%c)
n = 5; % set order of butterworth filter
Fc = [2, 10]; % set pass band frequencies
Wn = Fc/Fn; % get the normalised pass band frequencies
[b,a]=butter(n,Wn); % filter constants
ecg3 = filtfilt(b, a, ecg2); % filter
nexttile
plot(ecg3) % plot final filtered signal in time and frequency domains
xlabel('Samples (1 Sample = 1/1000s)')
ylabel('Amplitude (mV)')
title('Filtered ECG Signal')
nexttile
ecg3f = abs(fft(ecg3));
ecg3f = fftshift(ecg3f);
plot(f, ecg3f)
xlabel('Frequency (Hz)')
ylabel('Magnitude (dB)')
title('Filtered ECG Signal')
xlim([0, Fs/2])
```

Outputs:

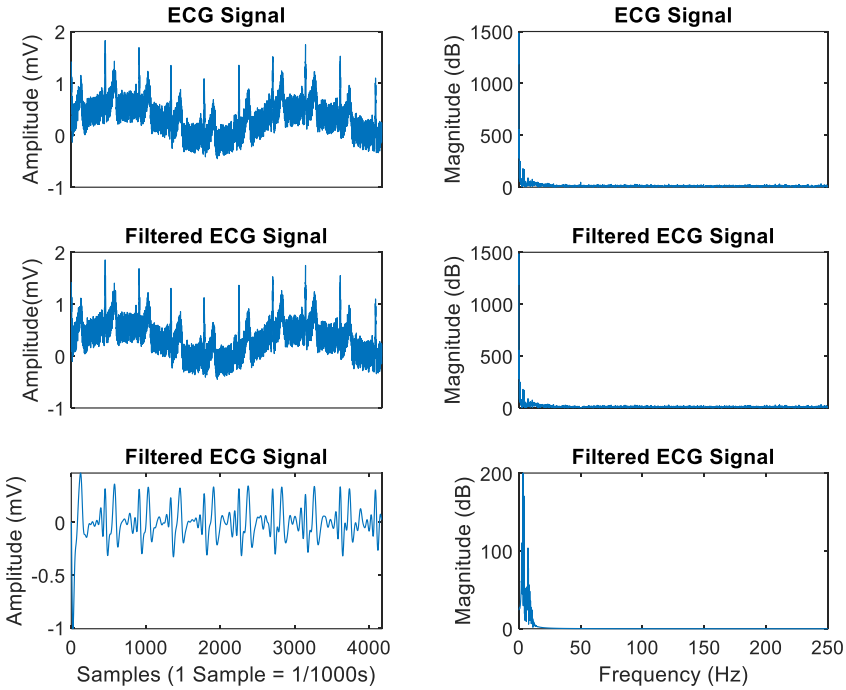

Figure 1

Discussion:

a)

The top left graph of Figure 1 show the noisy ecg signal in the time domain.

b)

The top right graph of Figure 1 show the noisy ecg signal in the frequency domain.

c)

The middle 2 graphs show the signal after the 50Hz interference was removed, figure 2 below is a zoomed in version of the frequencies for clarity, where it can be seen that the large peak at 50 Hz is not visible in the filtered signal

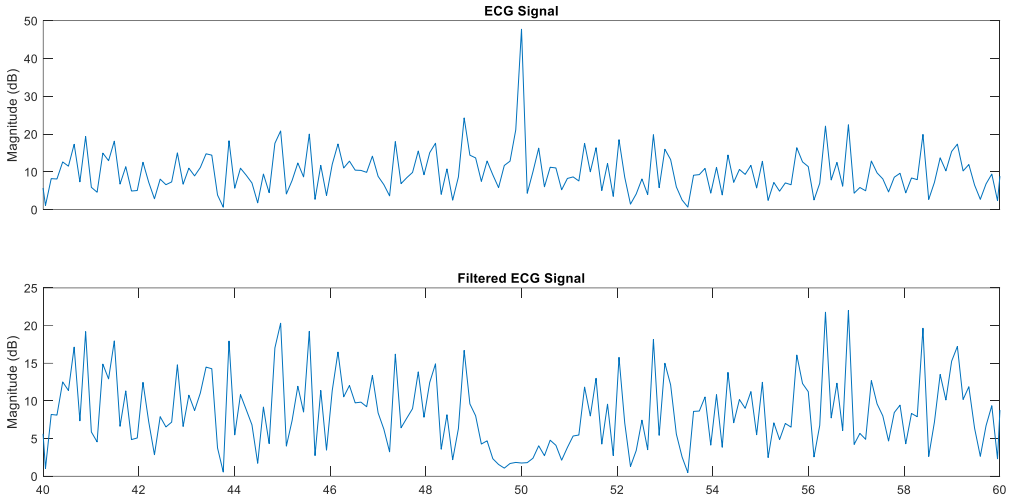

Figure 2

c)

The movement artefact, seen as a large sine wave had a period of approximately 4 seconds, this meant that it had a frequency of about 0.25Hz, a filter was used to remove this artefact, and ony other frequencies lower than 2Hz, an upper limit of 10 Hz was selected to remove the noise that was seen in the frequency graphs. The lower 2 graphs in Figure 1 show the final filtered ecg signal in the time and frequency domains.

Problem 3:

Code:

```
clear
clc
close all% clear the command windows, workspace and figures
load("G:\My Drive\engineering_work\Biosigs\Lab 4\noisy_EMG.mat")% load the EMG file

Fs = 1000; % smapling frequency
Fn = Fs/2; % nyquist frequency
Fint = 50; % the frequency of interference (power line)
len = length(noisy_EMG); % number of samples
tiledlayout(5,2) % tiled layout
%1)
nexttile % plot the noisy emg in the frequency and time domain
plot(noisy_EMG)
title('Noisy EMG signal')
ylabel('Amplitude (mV)')
xticks([])
noisy_EMGf = abs(fft(noisy_EMG));
noisy_EMGf = fftshift(noisy_EMGf);
f = (-len:2:len-1)*Fn/len;
nexttile
plot(f, noisy_EMGf)
title('Noisy EMG Frequencies')
ylabel('Magnitude (dB)')
xticks([])
xlim([0, Fs])
%2)
Fc = Fint;% notch filter routine for power line interference at 50 Hz
wo = Fc/Fn;
bw = wo/30;
[b,a] = iirnotch(wo,bw);
middenemg = filtfilt(b, a, noisy_EMG);
nexttile % plot emg without 50 Hz interference in time and frequency domain
plot(middenemg)
xticks([])
title('Noisy EMG signal, w/o 50Hz interference')
ylabel('Amplitude (mV)')
nexttile
fmiddenemg = abs(fft(middenemg));
fmiddenemg = fftshift(fmiddenemg);
plot(f, fmiddenemg)
title('Noisy EMG w/o 50Hz interference Frequencies')
ylabel('Magnitude (dB)')
xticks([])
xlim([0, Fs])
n = 5; %choose order of butterworth filter
Fc = [20, 450]; % choose range of useful frequencies
Wn = Fc/Fn;% calculate the normalised frequency
[b,a]=butter(n,Wn, 'stop');  %butterworth band stop filter
ignorefreqs = filtfilt(b, a, middenemg); % filter the useful parts out
clean_emg = middenemg - ignorefreqs*(.7);%generate the clean emg by scaling and subtracting
the ignored frequencies from the noisy emg
nexttile% plot clean emg in the time and frequency domains
plot(clean_emg)
xticks([])
title('Clean EMG Signal')
ylabel('Amplitude (mV)')
nexttile
fclean_emg = abs(fft(clean_emg));
fclean_emg = fftshift(fclean_emg);
plot(f, fclean_emg)
title('Clean EMG Frequencies')
ylabel('Magnitude (dB)')
xticks([])
xlim([0, Fs])
%3)
absemg = abs(clean_emg);%get the absolute value of the emg
nexttile% plot the absolute emg in the time and frequency domains
plot(absemg);
xticks([])
title('EMG Envelopes')
ylabel('Amplitude (mV)')
nexttile
```

```
fabsemg = abs(fft(absemg));
fabsemg = fftshift(fabsemg);
plot(f, fabsemg)
title('EMG Envelopes Frequencies')
ylabel('Magnitude (dB)')
xticks([])
xlim([0, Fs])
Fc = .8;%pick frequencies to filter the envelopes of the signal
Wn = Fc/Fn;%normalise
[b,a]=butter(n,Wn); %fitler constants
emgenv = filtfilt(b, a, absemg);%filter and obtain envelopes
%4) - 6)
gorlthresh = zeros(len,1);%initialise a zeros array for whether there is a signal or not
n = 3;%number of standard deviations from mean for cut off frequency
avg5 = mean(emgenv(1:Fs*5)); % find the mean of the first 5 seconds of just noise
sd5 = std(emgenv(1:Fs*5),1); % find the standard deviation of the first 5 seconds of noise
threshold = avg5+n*sd5; % Calculate the amplitude threshold = mean + n*standard deviation
for i = 1:len-1 %for loop to set the gorlthresh array equal to 1 if the signal is greater than
the threshold
    if emgenv(i)>threshold
        gorlthresh(i) = 1;
    end
end
nexttile%plot the envelopes in the time and freuqency domain, as well as whether or not the
signal is active on the time domain
yyaxis left
plot(emgenv)
title('EMG Envelopes and Activation Signal')
xlabel('Time (samples)')
ylabel('Amplitude (mV)')
hold on
yyaxis right
ylabel('Activated (0/1)')
ylim([0, 1.1])
plot(gorlthresh)

nexttile
femgenv = abs(fft(emgenv));
femgenv = fftshift(femgenv);
plot(f, femgenv)
title('EMG Envelopes and Activation Signal Frequencies')
xlabel('Frequecny (Hz)')
ylabel('Magnitude')
xlim([0, Fc])

insts = find((diff(gorlthresh))>0)/Fs;
insts
```

Output:

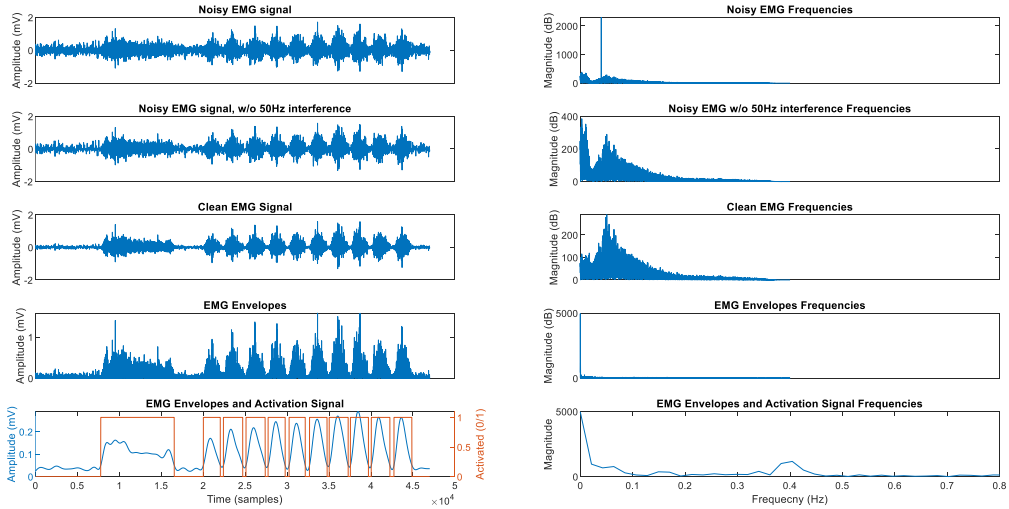

Figure 3

```
insts =

    7.7840

   20.0160

   22.4050

   25.1010

   27.7330

   30.2350

   32.6710

   35.0160

   37.5430

   40.0350

   42.7370
```

Discussion

1)

The noisy signal can be seen plotted in the frequency and time domains in the top graphs of Figure 3.

2)

The 50Hz interference was filtered out in the same method as it was filtered in the previous question, with a notch filter. The remaining frequencies outside the 20 – 450 Hz range were filtered out on their own, and subtracted from the main signal, with a reduced value, in order to minimise their effect on the signal, while maintaining some of their data. The “clean” signal is plotted in the third row of Figure 3.

3)

The steps needed to extract the envelopes was to :

1. Pre-process the signal – this was done in step 2
2. Find the modulus of the signal – this was done using the abs() function
3. Use a low pass filter on it – a cutoff frequency of 0.8 Hz was selected as it hade the signal adequately without noise

4)

The calculated mean for the noise of the first 5 seconds was 0.0373, and the standard deviation was 0.0044.

5)

After some trial and error, 2 standard deviations were found to be too sensitive, so 3 standard deviations were selected, this meant that there was a 97.5% that the noise values would not exceed this, value and that any big values that were recorded were not noise.

The bottom graphs in Figure 3 show the activations in red.

6)

The 11 activations were found using the diff() function in conjunction with the find() function, they can be seen below Figure3 above.